

Research Statement

I am Chang Jin. I am currently pursuing a master's degree in Computer Science and Technology at Tongji University in Shanghai, China, and I spent a semester as an exchange student at EPFL during my graduate studies. Through my academic training and research experience, my research interests lie in large language models' reasoning, robustness and safety. I have developed a strong passion for advancing the development of more reliable and trustworthy AI systems.

My motivation for pursuing this direction stems from my firsthand experience applying LLMs to complex, knowledge-intensive tasks. During my master's studies, I worked extensively on AI4Science problems, particularly drug discovery tasks such as molecular property prediction with large language models (LLM). In this work, I repeatedly encountered a fundamental challenge: even the most advanced models often hallucinate, generate inconsistent reasoning chains, and confidently provide incorrect answers. What struck me most was how these issues became critical in high-stakes scenarios, where a single unreliable output could mislead scientific analysis. For example, I found that many of the LLMs' analyses were chemically implausible, such as incorrectly identifying functional groups, miscounting carbon atoms, or producing predictions inconsistent with their own chain-of-thought reasoning. These experiences made me realize that improving task performance alone is insufficient. Unless we address the core issues of reliability, LLMs will remain unsafe for real-world deployment. This realization has since driven my passion for pursuing a PhD focused on LLM robustness and safety, ultimately aiming to enable more reliable and trustworthy models to support scientific discovery and decision-making in critical domains.

My previous research has consistently focused on improving the reliability of LLMs. I have explored reinforcement learning-based approaches to enhance LLM reasoning and abstention capabilities, enabling models to abstain more accurately from ambiguous or unanswerable questions. I also analyzed how decoding constraints influence model behavior. Currently, I am studying Chain-of-Thought monitoring techniques to identify illogical or inconsistent reasoning processes. In addition, my experience in AI4Science has taught me how to apply robust AI techniques to real-world scientific challenges. These projects have equipped me with strong technical foundations, including proficiency in Python and experience with modern deep learning frameworks, as well as solid research skills in experimental design, model analysis, and reliability evaluation, all of which I believe will allow me to contribute to my further research.

Here is a preliminary plan for my doctoral research. In my doctoral research, I aim to advance the safety and robustness of LLMs through several complementary directions. First, I plan to explore methods for knowledge injection, such as retrieval-augmented

generation and knowledge graph integration, to equip LLMs with more reliable and comprehensive knowledge that supports trustworthy reasoning. Second, building on my prior work on reinforcement learning for reasoning improvement, I plan to further enhance models' reasoning capabilities, enabling models to solve complex tasks more reliably and abstain when faced with uncertain or harmful queries. Third, I intend to develop fact-checking and verification pipelines that integrate causal reasoning to ensure logical consistency, along with monitoring mechanisms to detect and correct inconsistent or unsafe outputs. Last but not least, I aim to contribute to the development of robust evaluation frameworks that incorporate functional variants of benchmarks and metrics to provide a more accurate assessment of LLM robustness. Together, these directions aim to build LLMs with more robust and trustworthy behaviors and align them with real-world knowledge and human values.

Ultimately, my long-term goal is to become a researcher who advances the safety and reliability of large-scale learning systems such as large language models. I am particularly motivated by the potential of trustworthy AI to support evidence-based policy, accelerate scientific breakthroughs, and ensure equitable and safe deployment in real-world settings. I would be honored to pursue my doctoral studies to build the next generation of AI systems that are reliable, explainable, and aligned with human values.

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